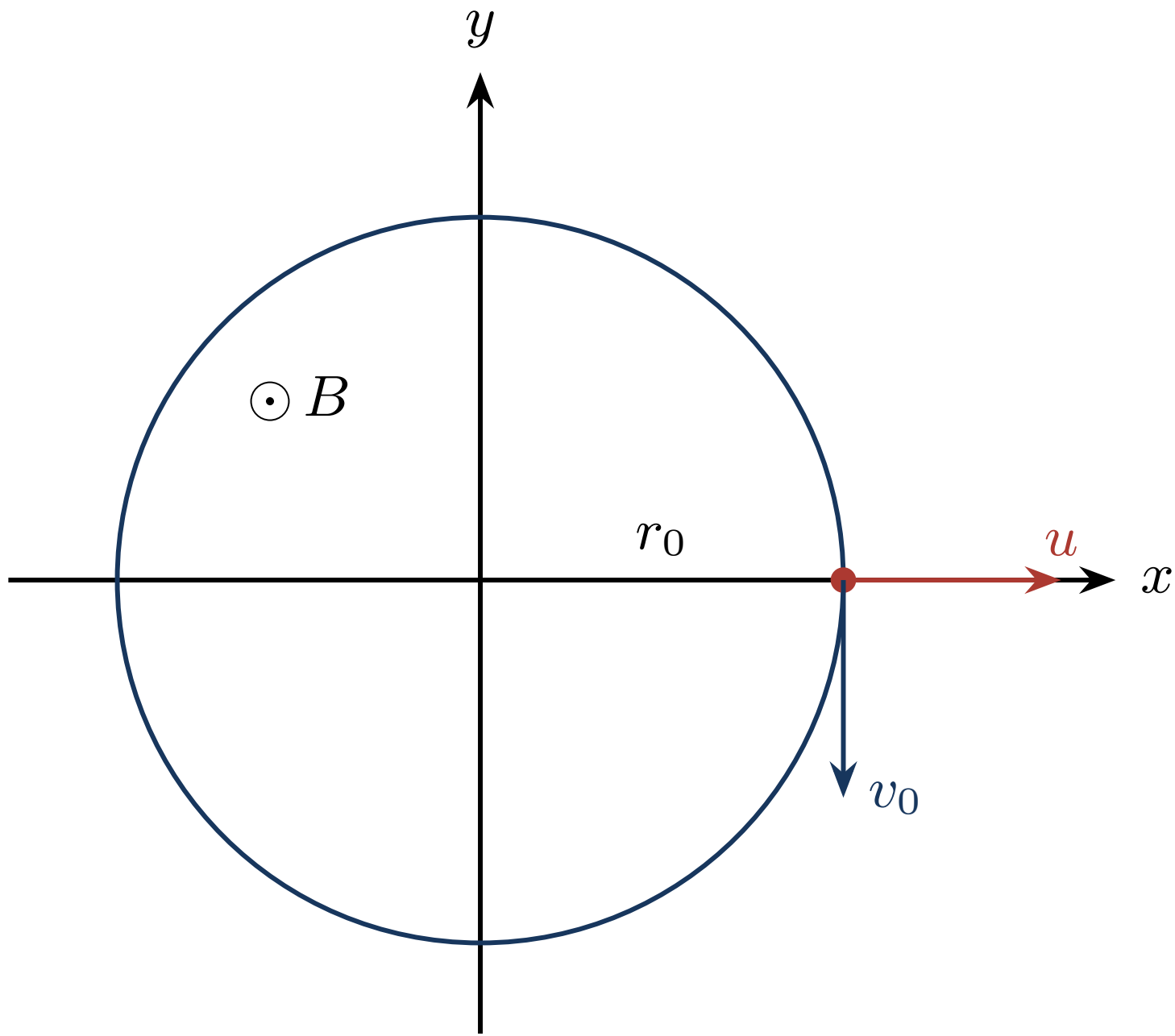
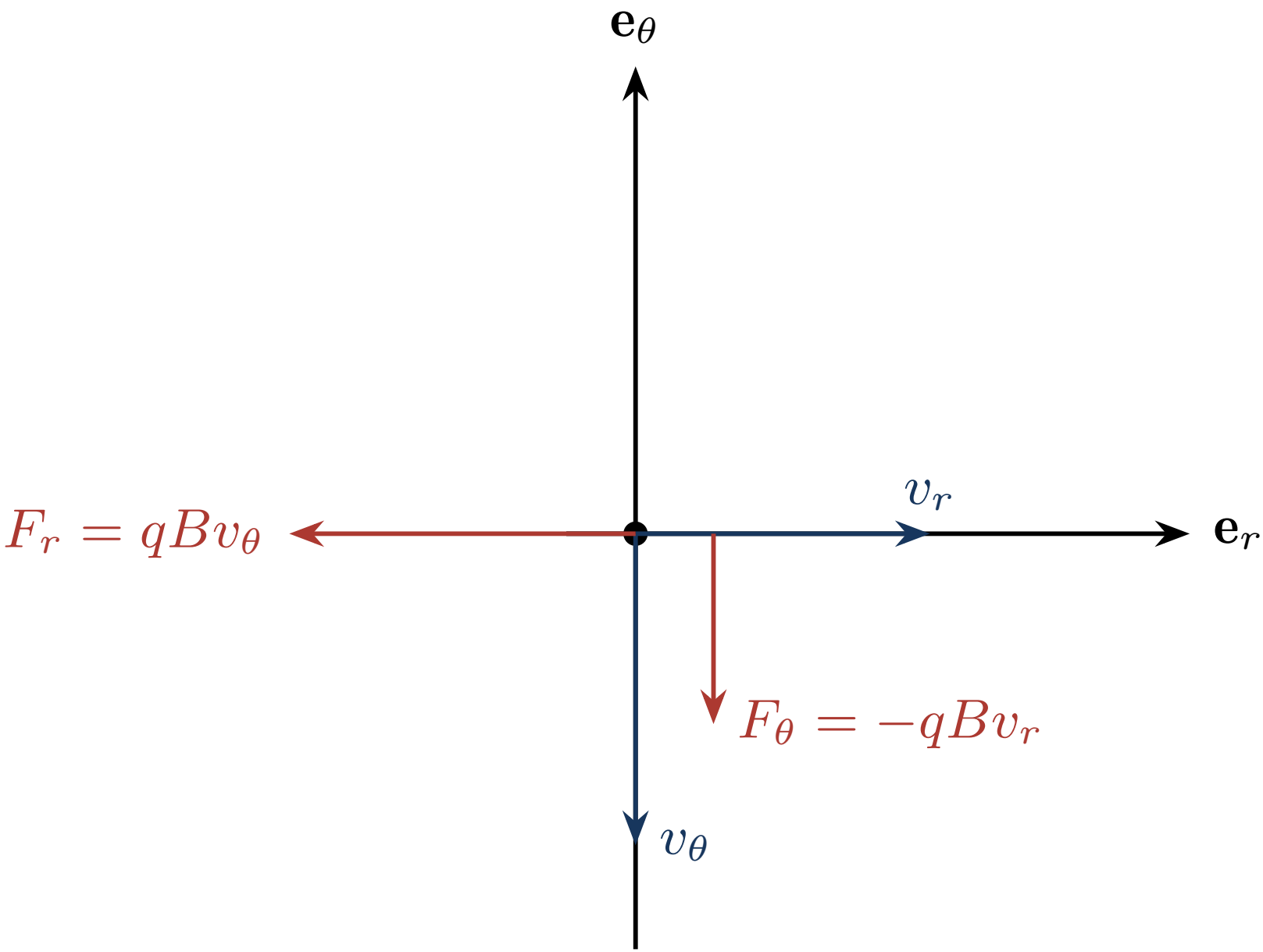
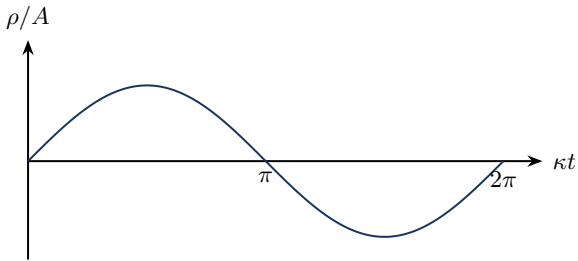
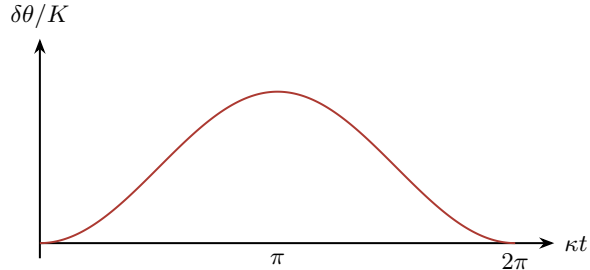


取 $q > 0$、磁场朝纸外，圆周运动顺时针；瞬时给予径向外冲量。

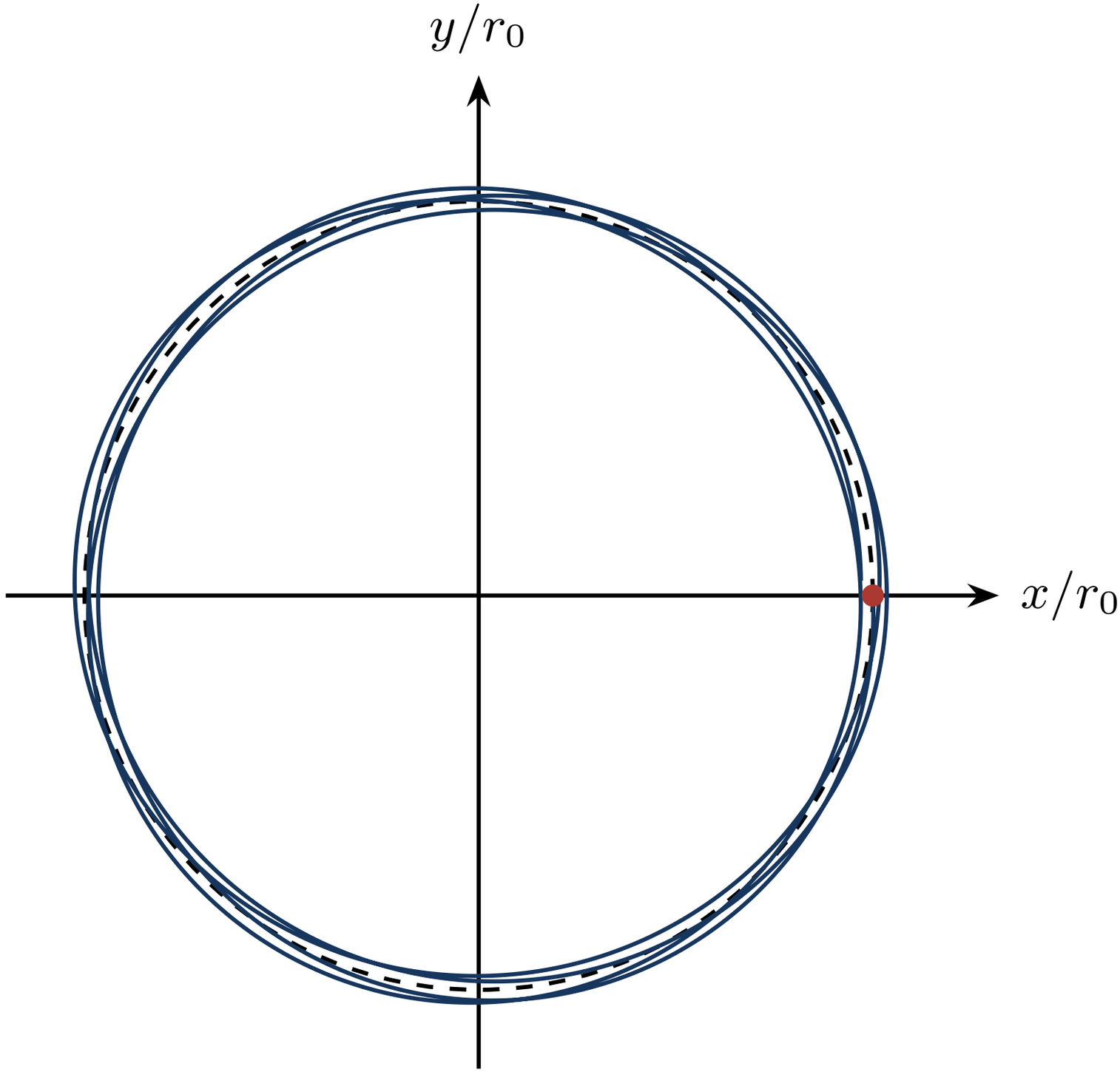


这里是粒子处的局部正交基；$\mathbf{e}_\theta$ 朝逆时针。机械角动量不守恒。

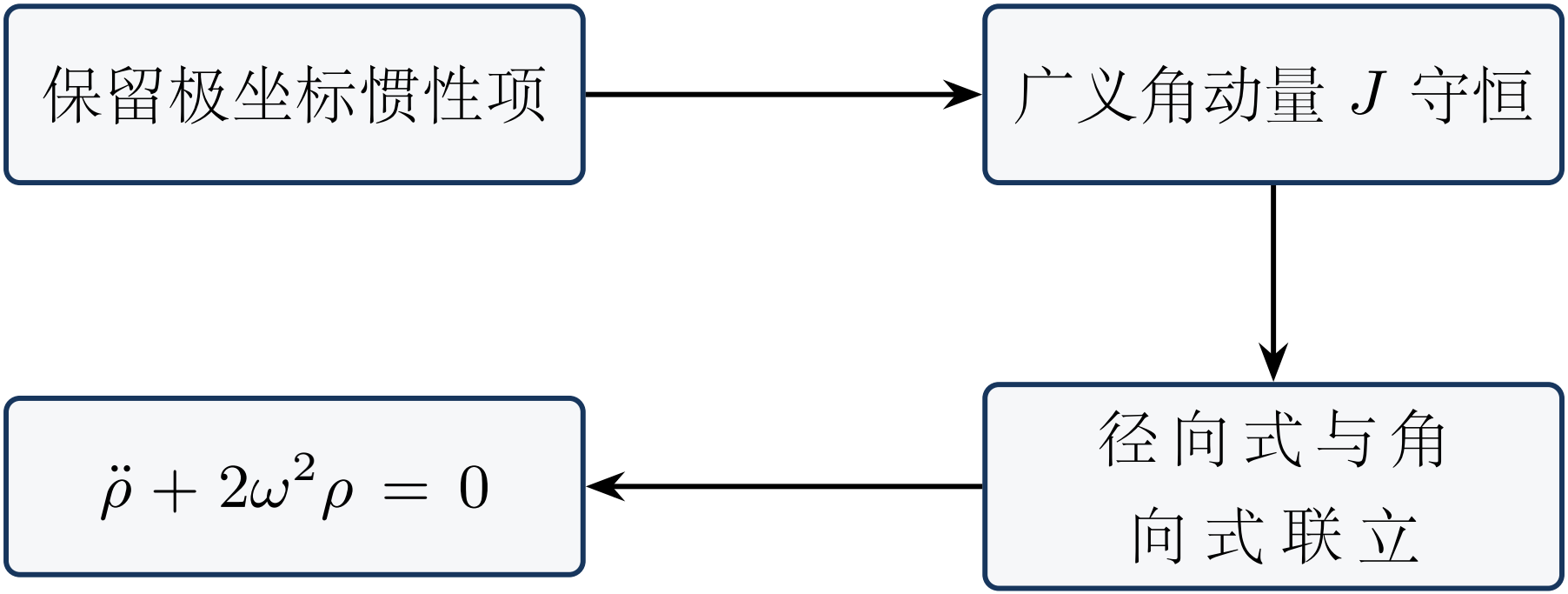




$\kappa = \sqrt{2}\omega,\ A = u/\kappa,\ K = u/(2\omega r_0)$；纯径向冲量要求 $\rho(0) = 0$。



示例 $u/v_0 = 0.05$，绘至 $\omega t = 8\pi$。这是线性近似轨迹，径向周期与转动周期不同。



径向振动频率 $\sqrt{2}\omega$；只用瞬时圆轨道条件会漏掉耦合项。